



CHEMICAL REACTOR DIGITAL TWIN

OPEN SOURCE

Vendor-neutral batch, CSTR and PFR R&D reference with calibration and validation workflows

MIT • Level 0 reference

Turn declared reaction and experiment data into an inspectable reactor-modeling workflow.

Validate units, simulate ideal reactors, calibrate kinetics, hold out validation data, quantify uncertainty, then optimize within reviewed constraints.

REACTOR BASELINE

Batch • CSTR • PFR
Mass + energy balances

CALIBRATABLE KINETICS

Arrhenius networks
Estimation + identifiability

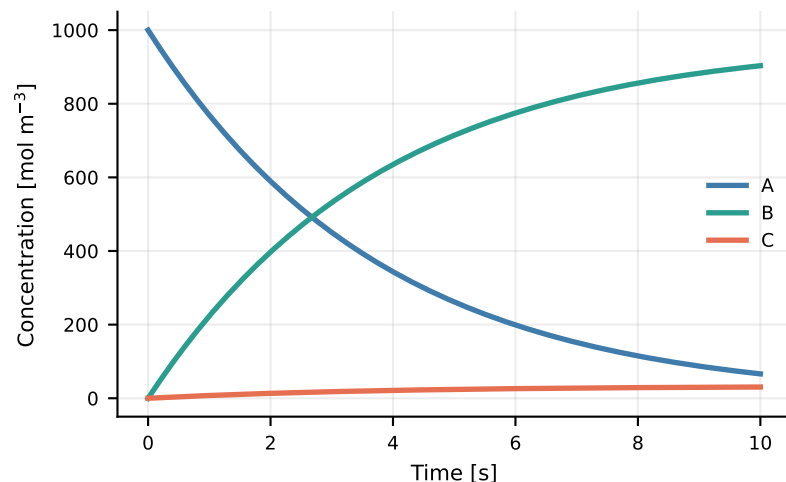
VALIDATION READY

Conservation • limits
Held-out metrics + uncertainty

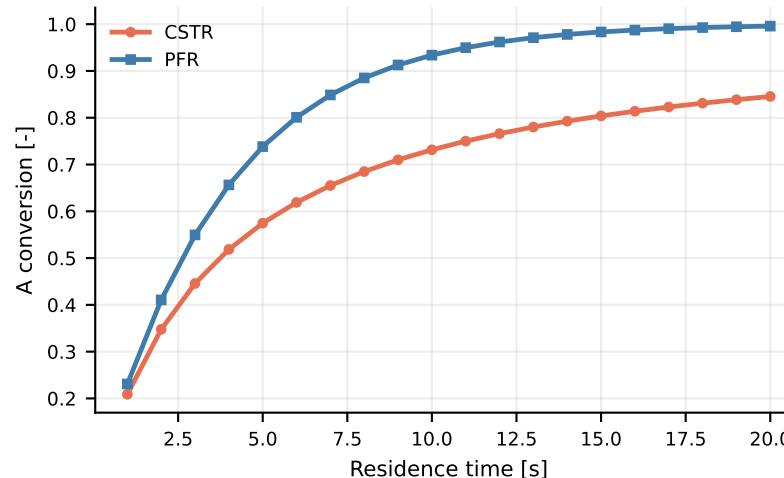
SAFETY BOUNDED

Solver diagnostics
Hypotheses, not procedures

Synthetic batch concentration response



Residence-time response at 330 K feed



0.934 conversion

10 s synthetic batch reference

0.967 B selectivity

Parallel-path synthetic network

334.36 K

Computed peak lumped temperature

2.68e-09 error

First-order analytical check

AN INSPECTABLE REACTION-ENGINEERING WORKFLOW

DECLARE

network + units

VALIDATE

schemas + gates

SIMULATE

balances + diagnostics

CALIBRATE

fit + hold-out check

ANALYZE

uncertainty + hypotheses